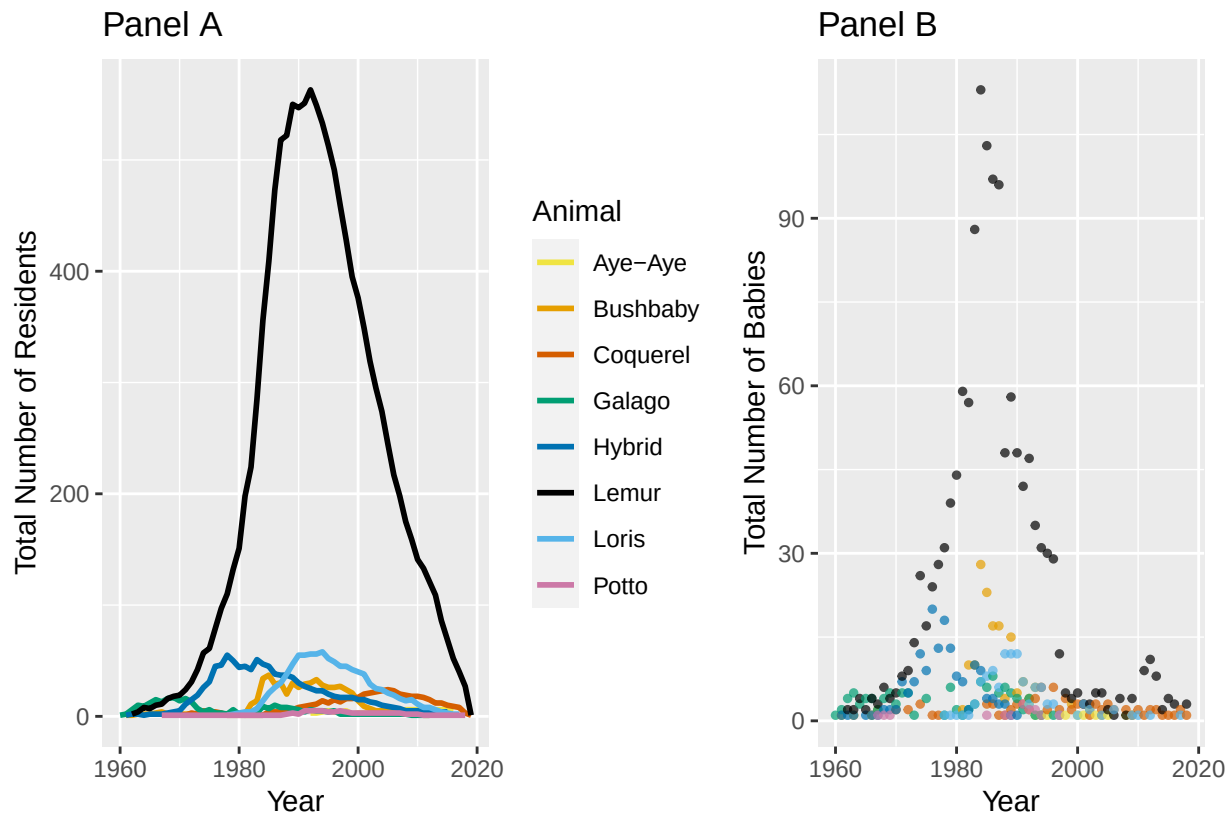


Assignment 4

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Prompt Questions: How has the number of total residents at the duke lemur center changed over time by primate type? How has the number of babies born at the duke lemur center changed over time by primate type?



The figures above show the total number of residents present and the total number of babies born each year by primate type at the Duke Lemur Center. Panel A shows that overall, Lemurs have had the most yearly residents and had a max in the early 1990s. Panel B shows that yearly, lemurs have had the most babies and had a max in the mid 1980s. One thing to note is that Panel A and B have different scales on the y-axis. Panel A has a max of 600 while Panel B has a max of 120. All the colors correspond with the same animal on both plots.

```

library(readxl)
DLCLH <- read_excel("~/Downloads/Duke Lemur Center Life History.xlsx")

library(dplyr)
library(ggplot2)
library(reshape2)
library(lubridate)
library(tidyr)
library(patchwork)

#this is filtering for only born at duke lemur
Born_Duke <- DLCLH %>%
  filter(Birth_Institution == 'Duke Lemur Center')

#This is seperating out years
Born_Duke <- Born_Duke %>%
  mutate(birthdate = as.POSIXct(DOB, format = "%Y-%m-%d")) %>%
  mutate(Year_Birth = year(birthdate)) %>%
  mutate(Month_Birth = month(birthdate)) %>%
  mutate(Day_Birth = day(birthdate))

#ditto
Born_Duke <- Born_Duke %>%
  mutate(deathdate = as.POSIXct(DOD, format = "%Y-%m-%d")) %>%
  mutate(Year_Death = year(deathdate)) %>%
  mutate(Month_Death = month(deathdate)) %>%
  mutate(Day_Death = day(deathdate))

#this is to check for na
unique(Born_Duke$Year_Birth)
unique(Born_Duke$Year_Death)

#filtering out the na
Born_Duke <- Born_Duke %>% filter(!is.na(Year_Birth))
Born_Duke <- Born_Duke %>% filter(!is.na(Year_Death))

#making sure that worked
unique(Born_Duke$Year_Birth)
unique(Born_Duke$Year_Death)

#This code is writing out and seperating taxon into specific groups
mutated_born <- Born_Duke %>%
  mutate(Animal = case_when(Taxon == "OGG" ~ "Galago",
    Taxon == "ECOL" ~ "Lemur",
    Taxon == "EUL" ~ "Hybrid",
    Taxon == "EFUL" ~ "Lemur",
    Taxon == "ERUF" ~ "Lemur",
    Taxon == "HGG" ~ "Lemur",
    Taxon == "VRUB" ~ "Lemur",
    Taxon == "CMED" ~ "Lemur",
    Taxon == "PPOT" ~ "Potto",
    Taxon == "GMOH" ~ "Bushbaby",
    Taxon == "EMON" ~ "Lemur",
    Taxon == "VVV" ~ "Lemur",

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Taxon == "EMAC" ~ "Lemur",
Taxon == "LCAT" ~ "Lemur",
Taxon == "EALB" ~ "Lemur",
Taxon == "VAR" ~ "Hybrid",
Taxon == "PCOQ" ~ "Coquerel",
Taxon == "MMUR" ~ "Lemur",
Taxon == "LTAR" ~ "Loris",
Taxon == "NCOU" ~ "Loris",
Taxon == "NPYG" ~ "Loris",
Taxon == "MZAZ" ~ "Lemur",
Taxon == "ECOR" ~ "Lemur",
Taxon == "ERUB" ~ "Lemur",
Taxon == "ESAN" ~ "Lemur",
Taxon == "EFLA" ~ "Lemur",
Taxon == "DMAD" ~ "Aye-Aye"

))

#This is making a new dataframe with new birth years and animal and tallying those
Birth_duke <- mutated_born %>%
  group_by(Year_Birth, Animal)%>%
  tally()

Birth_duke <- Birth_duke %>% filter(!is.na(Year_Birth))

unique(Birth_duke$Animal)

#this is the beginning of the present year lemur count
Dif_Years <- mutated_born %>%
  select(Animal, Year_Birth, Year_Death)

y = data.frame(yearpres=0)

for( i in 1:length(Dif_Years$Year_Birth)){
  x = data.frame(yearpres = rep(Dif_Years$Year_Birth[i]:Dif_Years$Year_Death[i]))
  y = rbind(y,x)
}

Total_Residents <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs")

#----- these are to separate chunks
Num_Galago <- Dif_Years %>%
  filter(Animal == "Galago")

y = data.frame(yearpres=0)

for( i in 1:length(Num_Galago$Year_Birth)){
  x = data.frame(yearpres = rep(Num_Galago$Year_Birth[i]:Num_Galago$Year_Death[i]))
  y = rbind(y,x)
}

Total_Galago <- y %>%
  filter(y!= 0) %>%

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group_by(yearpres) %>%
tally(name = "numberoflemurs") %>%
mutate(Animal = "Galago")
#-----
Num_Lemur <- Dif_Years %>%
  filter(Animal == "Lemur")

y = data.frame(yearpres=0)

for( i in 1:length(Num_Lemur$Year_Birth)){
  x = data.frame(yearpres = rep(Num_Lemur$Year_Birth[i]:Num_Lemur$Year_Death[i]))
  y = rbind(y,x)
}

Total_Lemur <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Lemur")
#-----
Num_Hybrid <- Dif_Years %>%
  filter(Animal == "Hybrid")

y = data.frame(yearpres=0)

for( i in 1:length(Num_Hybrid$Year_Birth)){
  x = data.frame(yearpres = rep(Num_Hybrid$Year_Birth[i]:Num_Hybrid$Year_Death[i]))
  y = rbind(y,x)
}

Total_Hybrid <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Hybrid")
#-----
Num_Potto <- Dif_Years %>%
  filter(Animal == "Potto")

y = data.frame(yearpres=0)

for( i in 1:length(Num_Potto$Year_Birth)){
  x = data.frame(yearpres = rep(Num_Potto$Year_Birth[i]:Num_Potto$Year_Death[i]))
  y = rbind(y,x)
}

Total_Potto <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Potto")

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#-----
Num_Bushbaby <- Dif_Years %>%
  filter(Animal == "Bushbaby")

y = data.frame(yearpres=0)

for( i in 1:length(Num_Bushbaby$Year_Birth)){
  x = data.frame(yearpres = rep(Num_Bushbaby$Year_Birth[i]:Num_Bushbaby$Year_Death[i]))
  y = rbind(y,x)
}

Total_Bushbaby <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Bushbaby")

#-----
Num_Coquerel <- Dif_Years %>%
  filter(Animal == "Coquerel")

y = data.frame(yearpres=0)

for( i in 1:length(Num_Coquerel$Year_Birth)){
  x = data.frame(yearpres = rep(Num_Coquerel$Year_Birth[i]:Num_Coquerel$Year_Death[i]))
  y = rbind(y,x)
}

Total_Coquerel <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Coquerel")

#-----
Num_Loris <- Dif_Years %>%
  filter(Animal == "Loris")

y = data.frame(yearpres=0)

for( i in 1:length(Num_Loris$Year_Birth)){
  x = data.frame(yearpres = rep(Num_Loris$Year_Birth[i]:Num_Loris$Year_Death[i]))
  y = rbind(y,x)
}

Total_Loris <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Loris")

#-----

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Num_AyeAye <- Dif_Years %>%
  filter(Animal == "Aye-Aye")

y = data.frame(yearpres=0)

for( i in 1:length(Num_AyeAye$Year_Birth)){
  x = data.frame(yearpres = rep(Num_AyeAye$Year_Birth[i]:Num_AyeAye$Year_Death[i]))
  y = rbind(y,x)
}

Total_AyeAye <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Aye-Aye")
#-----

PrimatebyYear = rbind(Total_AyeAye, Total_Bushbaby, Total_Coquerel, Total_Galago,
  Total_Hybrid, Total_Lemur, Total_Loris, Total_Potto)

#This is the code to make both into one chart
#It honestly is too busy and so much data is covered up
ggplot() +
  geom_line(data = Birth_duke, aes(Year_Birth, n, col = Animal), size = 1)+
  geom_point(data = PrimatebyYear, aes(yearpres, numberoflemurs, col = Animal),
    size = 1, alpha = .7)+
  scale_color_manual(values = c("#F0E442", "#E69F00", "#D55E00", "#009E73",
    "#0072B2", "black", "#56B4E9", "#CC79A7"))+
  labs(x = "Year", y = "Number of Lemurs")

#----- didn't use ^^

#This is the code for making two seperate graphs
#This is the plot for question 1
Q1 <- ggplot(PrimatebyYear, aes(x= yearpres, y = numberoflemurs, col = Animal)) +
  geom_line(size = 1) +
  scale_color_manual(values = c("#F0E442", "#E69F00", "#D55E00", "#009E73",
    "#0072B2", "black", "#56B4E9", "#CC79A7"))+
  labs(x = "Year", y = "Total Number of Residents", title = "Panel A")

#This plot answers question 2
Q2 <- ggplot(Birth_duke, aes(x= Year_Birth, y = n, col = Animal)) +
  geom_point(size = 1, alpha = .7) +
  scale_color_manual(values = c("#F0E442", "#E69F00", "#D55E00", "#009E73",
    "#0072B2", "black", "#56B4E9", "#CC79A7"))+
  labs(x = "Year", y = "Total Number of Babies", title = "Panel B")+
  theme(legend.position = "none")

Q1+Q2

```